

EquiLearn

Product Requirements Document

Core AI/ML & Accessibility Intelligence

Version	1.0
Scope	Core AI/ML only
Team	3 Members
Primary Goal	Multimodal educational-content transformation for accessibility

Scope note: Frontend, backend, APIs, database, authentication, deployment, website/app UI, and user-account management are deliberately postponed.

1. Product Overview

EquiLearn is a multimodal accessibility intelligence system that takes educational content in different formats and transforms it into representations suitable for users with different accessibility needs.

- PDFs
- Scanned documents
- Images
- Diagrams
- Charts
- Audio
- Video
- Raw text

The system processes this content using multiple AI/ML models and produces accessibility-oriented outputs for:

- 1 Blind users
- 2 Deaf / Hard-of-Hearing users
- 3 Low-vision users
- 4 Users with dyslexia

Core problem: How can the same educational material be automatically understood and transformed into different accessible representations without losing its meaning?

Input → Extraction → AI Understanding → Transformation → Accessibility Decision → Accessible Representation

2. Scope

2.1 In Scope

- Document processing
- Image processing
- OCR
- Computer vision
- Vision-Language Models
- Speech recognition
- Audio processing
- NLP
- LLM summarization
- LLM simplification
- Readability analysis
- Key-term extraction
- Alt-text generation
- Transcript generation
- Caption generation

- Text-to-Speech preparation
- Accessibility-specific transformations
- AI evaluation
- Error handling
- Model integration
- Core AI pipeline

2.2 Out of Scope

- Frontend
- Backend
- APIs
- Database
- Authentication
- Deployment
- Website/app UI
- User-account management
- Production infrastructure

3. Core Problem

Educational content can contain several modalities simultaneously. A textbook PDF may contain normal text, scanned text, diagrams, images, charts, captions, and technical labels. A lecture may contain speech, video, slides, and visual explanations.

Information	Required For
What text says	Blind, low-vision, dyslexia
What images depict	Blind, low-vision
What audio contains	Deaf/Hard-of-Hearing
What text means and how complex it is	Dyslexia, low-vision

4. Core System Objective

The system must transform heterogeneous learning material into a structured accessibility package.

```

raw_text
alt_text
transcript
summary
complexity_score
key_terms
reading_time
recommended_representations

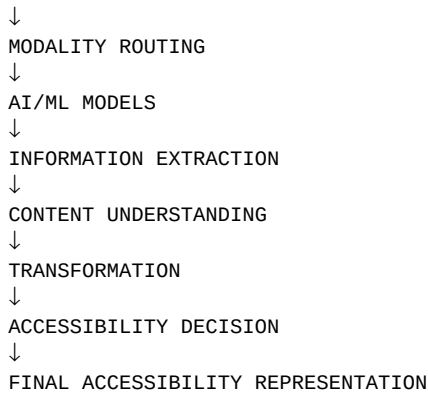
```

5. Core AI/ML Architecture

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RAW INPUT
(PDF / Image / Audio / Video / Text)
↓
DATA PREPARATION

```



The core system is not one single AI model. It is a routing graph in which different models handle different modalities.

6. Stage-by-Stage Product Workflow

STAGE 1 — INPUT ACQUISITION

Input: PDF, image, audio, video, or text.

Processing: Determine content type and extract raw components. This stage is basic file/format processing, not AI.

Output: Raw bytes, decoded streams, text content, images, audio, video.

STAGE 2 — DATA PREPARATION

Images: resize, denoise, deskew, normalize.

Audio: resample to 16 kHz mono, trim silence, normalize amplitude.

Video: extract audio track and keyframes where required.

Text: normalize Unicode, fix encoding, split into paragraphs.

STAGE 3 — MODALITY ROUTING

Digital PDF with usable text → direct text extraction.

Scanned page/image → OCR.

Speech → ASR/Whisper.

Visual content → VLM.

Text → NLP + LLM.

STAGE 4 — AI/ML PROCESSING

PaddleOCR for OCR; Tesseract as fallback; Qwen2.5-VL / GPT-4o-type VLM for visual understanding; Whisper large-v3 for speech recognition; GPT-4o / Gemini Flash / Llama for summarization and simplification; Web Speech API / gTTS for text-to-speech.

STAGE 5 — INFORMATION UNDERSTANDING

Text: readability, sentence length, lexical density, passive voice, complexity, key terms.

Images: objects, actions, spatial relationships, visual descriptions.

Audio: transcript segments, timestamps, speaker changes, topic shifts.

STAGE 6 — TRANSFORMATION

Text → summarization, simplification, key-term extraction.

Images → alt-text.

Audio/video → transcript, timestamped captions, VTT.

Text → speech-ready representation.

STAGE 7 — ACCESSIBILITY INTELLIGENCE

A rule-based decision layer selects the appropriate representation for the accessibility profile.

STAGE 8 — FINAL ACCESSIBILITY PACKAGE

All processed information is combined into a structured package containing raw text, alt-text, transcript, summary, complexity score, key terms, reading time, and recommended representations.

7. Member-by-Member Workflow

MEMBER 1 — Vision & Document AI

Primary responsibility: Documents + Images + OCR + Visual Understanding

M1-STAGE 1 — Document Processing

Understand PDF types, implement digital-text extraction, identify when OCR is required, handle scanned pages, and convert pages into images when needed.

M1-STAGE 2 — Image Preprocessing

Resize, denoise, deskew, normalize, and prepare images for OCR/VLM processing.

M1-STAGE 3 — OCR

Primary: PaddleOCR. **Fallback:** Tesseract.

Pipeline: Image → text detection → text recognition → bounding boxes → confidence.

Output: Text, bounding boxes, confidence scores.

M1-STAGE 4 — Vision-Language Model

Use Qwen2.5-VL-3B-Instruct or an equivalent approved VLM to understand objects, scenes, diagrams, spatial relationships, and visual content. OCR tells what text is written; VLM helps explain what the whole visual means.

M1-STAGE 5 — Alt-Text

Image → VLM → description → alt-text. For complex diagrams, combine OCR labels with VLM visual relationships to form a structured description.

M1-STAGE 6 — Blind User Support

PDF → text + images → OCR + VLM → text labels + visual description → Member 3 language processing → unified description.

M1-STAGE 7 — Low-Vision Support

Provide OCR text, image descriptions, visual information, and important visual content. Presentation enhancement itself is outside this core AI PRD.

M1-STAGE 8 — Evaluation

OCR: CER, confidence, recognition failures. VLM: human description rating, plus BLEU/ROUGE where appropriate, and hallucination cases.

Member 1 Final Deliverables

- PDF text-extraction workflow
- OCR module
- Tesseract fallback
- OCR confidence handling
- VLM image-understanding workflow
- Alt-text generation

- Diagram understanding
- Visual-information extraction
- OCR evaluation
- VLM evaluation

MEMBER 2 — Audio & Speech AI

Primary responsibility: Audio + Video Audio + Speech Recognition + Transcription + Caption Generation + TTS investigation

M2-STAGE 1 — Audio Extraction

For video: Video → FFmpeg → audio track. Then audio → 16 kHz → mono → silence trimming → Whisper-ready audio.

M2-STAGE 2 — Automatic Speech Recognition

Model: Whisper large-v3 / large-v3-turbo.

Pipeline: audio → mel-spectrogram → transformer encoder/decoder → text tokens → timestamps.

Output: transcript, segments, start/end timestamps, language.

M2-STAGE 3 — Deaf/Hard-of-Hearing Workflow

Lecture video → extract audio → preprocess → Whisper → speech-to-text → timestamped segments → VTT formatting → captions + transcript.

M2-STAGE 4 — Transcript Processing

Handle timestamp generation, segment creation, transcript formatting, speaker-change approximation, and technical terminology preservation.

M2-STAGE 5 — VTT Generation

Convert transcript + timestamps into WebVTT captions.

M2-STAGE 6 — TTS

Investigate Web Speech API and gTTS. Text → TTS → speech. TTS supports blind users, can be optional for low-vision users, and can assist dyslexia workflows.

M2-STAGE 7 — Audio Evaluation

Evaluate WER, timestamp accuracy, noisy audio, accented speech, incorrect transcription, and hallucinated speech.

Member 2 Final Deliverables

- Audio preprocessing
- Video → audio extraction
- Whisper pipeline
- Transcript generation
- Timestamp generation
- VTT generation
- Deaf/HoH accessibility workflow
- TTS investigation/component
- ASR error handling

- WER evaluation

MEMBER 3 — NLP & Accessibility Intelligence

Primary responsibility: Text Understanding + LLM + Simplification + Summarization + Dyslexia + Accessibility Decision Logic

M3-STAGE 1 — Text Processing

Process text from digital PDFs, OCR, visual descriptions, and transcripts. Perform paragraph/sentence segmentation, Unicode normalization, and complexity analysis.

M3-STAGE 2 — Readability Analysis

Calculate Flesch Reading Ease, Flesch-Kincaid Grade Level, average sentence length, lexical density, and word-frequency bands.

M3-STAGE 3 — Complexity Detection

Determine whether content is difficult, identify difficult sentences/words, and identify information that must remain unchanged.

M3-STAGE 4 — LLM Summarization

Use an approved LLM such as GPT-4o, Gemini Flash, or Llama for important-information identification and condensed representations.

M3-STAGE 5 — Dyslexia Simplification

Original text → complexity analysis → LLM → simplification → readability check. If readability is insufficient, refine the prompt and run again, with a maximum of two refinement iterations.

M3-STAGE 6 — What the LLM Changes

Syntactic simplification, lexical simplification, passive-to-active conversion where appropriate, and clearer coreference. The requirement is to reduce linguistic complexity without losing important meaning.

M3-STAGE 7 — Blind User Language Processing

Combine OCR text + VLM visual description + context → LLM → unified meaningful description.

M3-STAGE 8 — Accessibility Decision Logic

User profile → deterministic decision table → required representation. This is rule-based, not ML.

M3-STAGE 9 — Final Content Transformation

Combine outputs into summary, simplified text, key terms, complexity score, unified visual descriptions, TTS-ready sentences, and simplified transcripts as required.

M3-STAGE 10 — NLP/LLM Evaluation

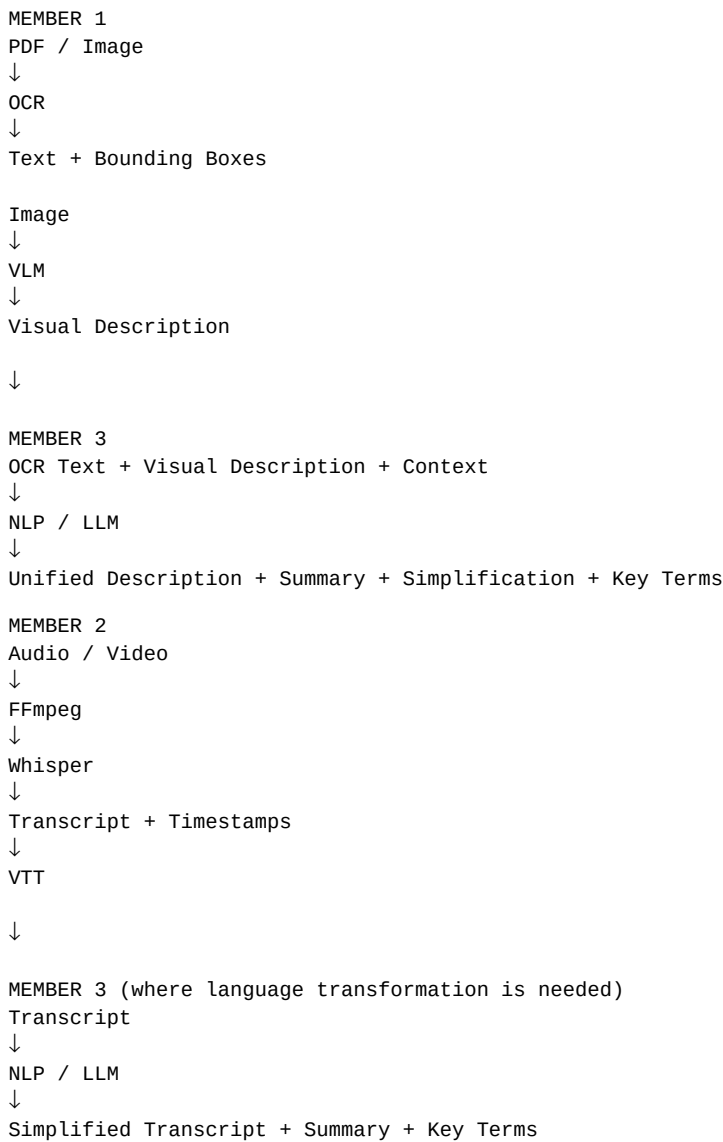
Summarization: ROUGE-L/BERTScore. Simplification: Flesch Reading Ease/FKGL/SARI. Human evaluation: semantic preservation, readability, hallucination.

Member 3 Final Deliverables

- Text-processing pipeline
- Readability analyzer
- Complexity analysis
- Key-term extraction

- LLM summarization
- LLM simplification
- Iterative refinement
- Dyslexia pipeline
- Blind-user description merging
- Accessibility decision rules
- Final transformation logic
- NLP/LLM evaluation

8. How the Three Members Connect



9. Shared Core

Component	Owner	Used By
OCR	Member 1	Blind, Low Vision
VLM	Member 1	Blind, Low Vision

Component	Owner	Used By
Whisper	Member 2	Deaf
LLM	Member 3	Blind, Low Vision, Dyslexia
TTS	Member 2 / Shared	Blind, Low Vision, Dyslexia
Accessibility rules	Member 3	All profiles

Shared modules should be implemented once and reused rather than duplicated.

10. Accessibility-Specific Workflows

10.1 Blind

```

PDF
↓
pypdf
↓
Text + Images
↓
OCR + VLM
↓
Text Labels + Visual Description
↓
LLM
↓
Unified Description
↓
TTS
↓
Audio Representation

```

10.2 Deaf / Hard-of-Hearing

```

Video
↓
FFmpeg
↓
Audio
↓
Whisper
↓
Transcript + timestamps
↓
VTT
↓
Captions + transcript

```

10.3 Low Vision

```

PDF / Image / Video
↓
OCR + VLM
↓
Text + Visual Description
↓
LLM
↓
Simplification + Summary
↓
Accessibility representation

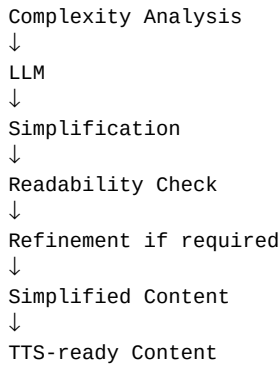
```

10.4 Dyslexia

```

Original Text
↓

```



11. Data and Training Requirements

Hackathon implementation: No model training is required. The major models are pretrained. The work focuses on integration, validation, evaluation, and transformation logic.

Task	Model	Training Required
OCR	PaddleOCR / Tesseract	No
Visual understanding	Qwen2.5-VL	No
ASR	Whisper	No
Summarization	GPT-4o / Gemini / Llama	No
Simplification	LLM	No
TTS	Web Speech / gTTS	No

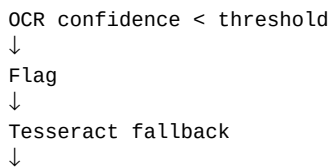
For validation/testing, the source plan proposes approximately 10 validation samples and 5 unseen test samples for major tasks.

12. Evaluation Plan

Member	Component	Metric
M1	OCR	CER
M1	VLM	Human rating + BLEU/ROUGE
M2	ASR	WER
M2	TTS	Intelligibility / MOS
M3	Summarization	ROUGE-L / BERTScore
M3	Simplification	FRE / FKGL / SARI
M3	Accessibility transformation	Human evaluation

13. Failure Handling

Member 1



VLM fallback where appropriate

Member 2

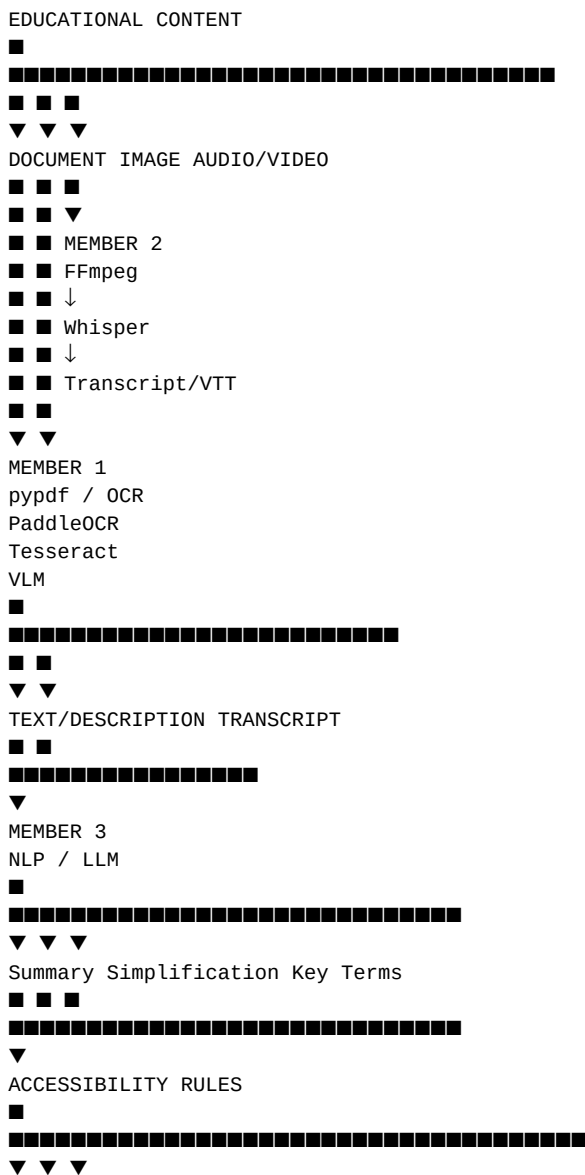
Low ASR confidence
↓
Flag uncertain segment
↓
Human review / fallback

Member 3

LLM summary loses information
↓
Key-term comparison
↓
Extractive fallback

Readability insufficient
↓
Refine prompt
↓
Run again

14. Final Team Architecture



BLIND DEAF LOW VISION



TTS Captions Simplified/

+ Alt-text + Transcript enhanced content



DYSLEXIA



Simplified + TTS

15. Member Ownership Summary

Member	Owns	Primary Profiles
Member 1 — Vision & Document Intelligence	PDF → document extraction → image preprocessing → OCR → VLM	Blind, Deaf, Low Vision
Member 2 — Audio & Speech Intelligence	Audio → preprocessing → Whisper → transcript → timestamps → VTT → TTS	Blind, Deaf, Low Vision, Dyslexia
Member 3 — NLP & Accessibility Intelligence	Text → NLP → complexity → LLM → summarization → simplification	Dyslexia, enhanced support for Blind, Low Vision, tra

16. Definition of Done

Member 1

- PDF/image content can be extracted
- OCR works
- OCR fallback exists
- VLM can describe selected images
- Alt-text is generated
- OCR/VLM outputs are evaluated

Member 2

- Audio can be extracted
- Whisper produces transcripts
- Timestamps are generated
- VTT can be produced
- TTS workflow is available/investigated
- WER is measured

Member 3

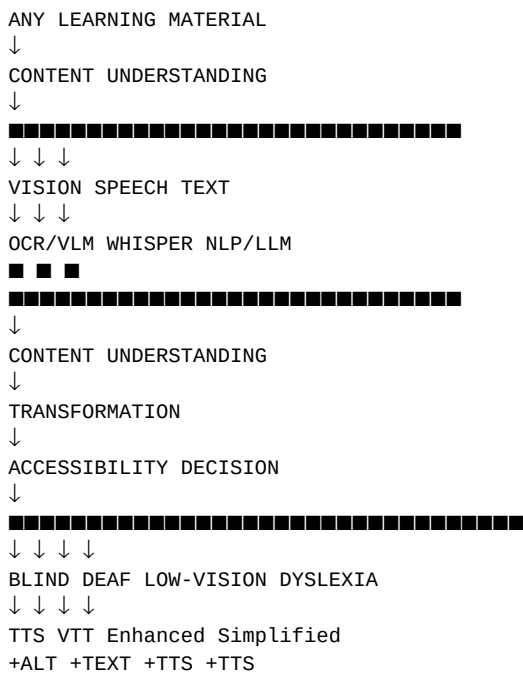
- Text complexity can be measured
- Summaries can be generated
- Text can be simplified
- Readability can be checked
- Simplification can be iteratively refined
- Key terms can be extracted
- Accessibility decision rules are implemented

- NLP/LLM outputs are evaluated

Team

- M1 outputs can be consumed by M3
- M2 outputs can be consumed by M3 where required
- Shared LLM workflow works
- Shared TTS workflow works
- Four accessibility profiles can be demonstrated
- Evaluation results are documented
- Failure cases are handled

17. Final Core AI Goal



Central principle: EquiLearn is not training one giant AI model. It combines pretrained multimodal models through a structured routing and transformation pipeline. The core intelligence is in how OCR, VLM, ASR, NLP/LLM, TTS, and deterministic accessibility rules work together.